Assignment 2A - SQL & R (Movie Ratings)

Ziaul Hoque

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Approach

This assignment 2A - SQL & R related to movie ratings, here movie ratings are collected from five people for six selected movies. Every person will rate the movies based on have seen, scale from 1 to 5. If any participant has not seen a movie, rating will be treated as missing rather than assigning an arbitrary value. So I will create a database name *"MovieRatings"* using GUI *pgAdmin 4* from *PostgreSQL* and will store all collected data into relevant three tables:

- users
- movies
- ratings

After creating and populating the database, I will use SQL queries to join the tables and retrieve the movie rating data. And then connect to *PostgreSQL database to R* and load the query results into a dataframe.

For Data Analysis part, I will inspect and clean the data, address any missing ratings through ***R*** language and use `dplyr` package to calculate simple summaries - such as the number of ratings and average rating for each movie.

Codebase

Create Database & related tables Following SQL code was used to create database & related tables.

```
CREATE DATABASE MovieRatings;

CREATE TABLE users (
    user_id SERIAL PRIMARY KEY,
    name VARCHAR(50)
);

CREATE TABLE movies (
    movie_id SERIAL PRIMARY KEY,
    title VARCHAR(100)
);

CREATE TABLE ratings (
    user_id INTEGER REFERENCES users(user_id),
    movie_id INTEGER REFERENCES movies(movie_id),
    rating INTEGER CHECK (rating BETWEEN 1 AND 5),
    PRIMARY KEY (user_id, movie_id)
);
```

```
INSERT INTO users (name)
VALUES
    ('Rudolf Schenker'),
    ('John Poli'),
    ('Robert Burnham'),
    ('Tracy Harrington'),
    ('Nancy Accorsi');

INSERT INTO movies (title)
VALUES
    ('Jason Bourne'),
    ('Pirates of the Caribbean'),
    ('Patriot'),
    ('Italian Job'),
    ('Robinhood & Sherwood Jungle'),
    ('Godzilla');
```

Store all records into table I collected and organized all relevant movie ratings data in Google Sheets, leaving cells blank when a participant had not seen or rated a movie. Then transferred all those data into PostgreSQL using three related tables for *users*, *movies*, and *ratings*. Each available rating was inserted into the ratings table using the corresponding user and movie IDs. Missing ratings were not inserted into the ratings table.

```
INSERT INTO ratings (user_id, movie_id, rating)
VALUES
    (1, 1, 4),
    (1, 2, 2),
    (1, 3, 3),
    (1, 4, 5),
    (1, 5, 4),
    (1, 6, 1),

    (2, 1, 4),
    (2, 2, 4),
    (2, 4, 4),
    (2, 5, 2),

    (3, 1, 1),
    (3, 2, 5),
    (3, 3, 2),
    (3, 5, 4),
    (3, 6, 4),

    (4, 1, 5),
    (4, 3, 5),
    (4, 4, 5),
    (4, 5, 4),
    (4, 6, 5),

    (5, 1, 3),
    (5, 2, 3),
    (5, 3, 2),
    (5, 4, 2),
    (5, 5, 3),
```

```
    (5, 6, 5);
```

Query Movie Ratings The three tables were then joined using the user and movie IDs to produce a readable dataset containing each participant, movie, and corresponding rating.

```
SELECT
    users.name,
    movies.title,
    ratings.rating
FROM ratings
JOIN users
    ON ratings.user_id = users.user_id
JOIN movies
    ON ratings.movie_id = movies.movie_id;
```

Load SQL Data into R The PostgreSQL database was connected directly to R using the `DBI` and `RPostgres` packages.

```
movie_ratings <- dbGetQuery(con, "
  SELECT
    users.name,
    movies.title,
    ratings.rating
  FROM users
  CROSS JOIN movies
  LEFT JOIN ratings
    ON users.user_id = ratings.user_id
    AND movies.movie_id = ratings.movie_id
")

movie_ratings
```

```
##                name                       title rating
## 1   Rudolf Schenker                Jason Bourne      4
## 2   Rudolf Schenker    Pirates of the Caribbean      2
## 3   Rudolf Schenker                     Patriot      3
## 4   Rudolf Schenker                 Italian Job      5
## 5   Rudolf Schenker Robinhood & Sherwood Jungle      4
## 6   Rudolf Schenker                    Godzilla      1
## 7         John Poli                Jason Bourne      4
## 8         John Poli    Pirates of the Caribbean      4
## 9         John Poli                     Patriot     NA
## 10        John Poli                 Italian Job      4
## 11        John Poli Robinhood & Sherwood Jungle      2
## 12        John Poli                    Godzilla     NA
## 13   Robert Burnham                Jason Bourne      1
## 14   Robert Burnham    Pirates of the Caribbean      5
## 15   Robert Burnham                     Patriot      2
## 16   Robert Burnham                 Italian Job     NA
## 17   Robert Burnham Robinhood & Sherwood Jungle      4
## 18   Robert Burnham                    Godzilla      4
## 19 Tracy Harrington                Jason Bourne      5
## 20 Tracy Harrington    Pirates of the Caribbean     NA
## 21 Tracy Harrington                     Patriot      5
```

```
## 22 Tracy Harrington                  Italian Job      5
## 23 Tracy Harrington Robinhood & Sherwood Jungle      4
## 24 Tracy Harrington                     Godzilla      5
## 25    Nancy Accorsi                 Jason Bourne      3
## 26    Nancy Accorsi    Pirates of the Caribbean      3
## 27    Nancy Accorsi                      Patriot      2
## 28    Nancy Accorsi                  Italian Job      2
## 29    Nancy Accorsi Robinhood & Sherwood Jungle      3
## 30    Nancy Accorsi                     Godzilla      5
```

Data Analysis At first, I inspect the data frame and checked for missing ratings. There are four missing ratings in the dataset. These missing values represent movies that participants did not rate. Rather than replacing the missing values with an artificial rating, I will exclude them when calculating summary statistics. This prevents unrated movies from affecting the average ratings.

```r
head(movie_ratings)
```

```
##              name                       title rating
## 1 Rudolf Schenker                Jason Bourne      4
## 2 Rudolf Schenker    Pirates of the Caribbean      2
## 3 Rudolf Schenker                     Patriot      3
## 4 Rudolf Schenker                  Italian Job      5
## 5 Rudolf Schenker Robinhood & Sherwood Jungle      4
## 6 Rudolf Schenker                     Godzilla      1
```

```r
str(movie_ratings)
```

```
## 'data.frame':    30 obs. of  3 variables:
##  $ name  : chr  "Rudolf Schenker" "Rudolf Schenker" "Rudolf Schenker" "Rudolf Schenker" ...
##  $ title : chr  "Jason Bourne" "Pirates of the Caribbean" "Patriot" "Italian Job" ...
##  $ rating: int  4 2 3 5 4 1 4 4 NA 4 ...
```

```r
sum(is.na(movie_ratings$rating))
```

```
## [1] 4
```

```r
movie_summary <- movie_ratings %>%
  group_by(title) %>%
  summarise(
    number_of_ratings = sum(!is.na(rating)),
    average_rating = mean(rating, na.rm = TRUE)
  ) %>%
  arrange(desc(average_rating))

movie_summary
```

```
## # A tibble: 6 x 3
##   title                       number_of_ratings average_rating
##   <chr>                                   <int>          <dbl>
## 1 Italian Job                                 4           4
## 2 Godzilla                                    4           3.75
## 3 Pirates of the Caribbean                    4           3.5
## 4 Jason Bourne                                5           3.4
## 5 Robinhood & Sherwood Jungle                 5           3.4
## 6 Patriot                                     4           3
```

The results show that *"Italian Job"* had the highest average rating at 4.00, while *"Patriot"* had the lowest

average rating at 3.00. *"Jason Bourne"* and *"Robinhood & Sherwood Jungle"* received ratings from all five participants, while the remaining movies each received four ratings. Missing ratings were excluded from the calculation of the averages.

Database connection close -

```
dbDisconnect(con)
```